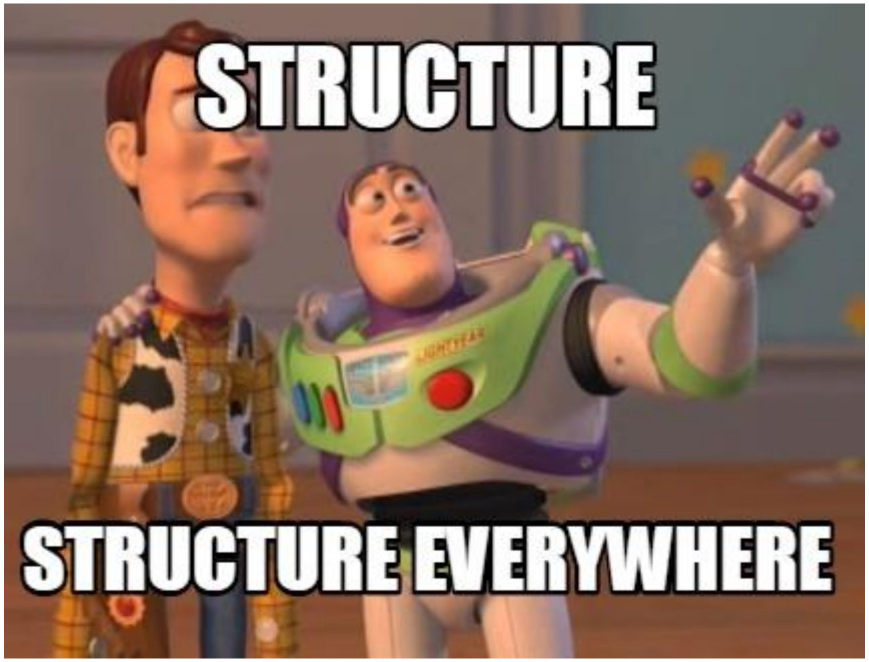
Education Session #4
Part 1: Frameworks continued



Note: You are all doing great. Take a deep breath & reward yourself with a DONUT!

Agenda: Last Week & This Week in Consulting

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Last Week:

- **Complete your resume -** MBA2 peer coaches are available to you. Resumes will be available to employers on 10/10 in RR
- **Odyssey & Behavioral Prep**
- **Case starts and frameworks**
- **Full case experience in your FACT group**

This Week:

- **Keep practicing fit + odyssey!**
- **Frameworks deep dive**
- **Practice going deeper with your insights on charts** (don't worry about timing for now)
- **You'll be doing a full interviewee-led case start with FACT groups,** don't be too hard on yourself but remember everything you've already learned!!

Feedback for us

98% of you found last week's session helpful!

What you loved:

- Live case examples
- Individual networking journeys
- Going step by step
- Easing the tension around casing

What you want to see:

- More, more, more live examples
- Thought process when we share live examples

Want more time from us? You got it!

Quick note from CDO

- Coffee chat registration will start at Noon (not 1 PM) going forward
- Companies are aware that interest in connecting with them is strong and they are going to provide MANY opportunities to connect with them between now and December
- Firms will always carve out a way to connect with someone who wasn't able to sign up
- Sign up for a max of 2 or 3 of the coffee chat options per firm to ensure equity in the process

Now, some feedback for you

DO NOT use your phone or laptop at an in-person event

- This is unprofessional and reflects poorly on you and your classmates
- If you need to use your cellphone to note down the firm rep's contact details, be respectful and inform the person that you are making a note

It is your responsibility to set up calendar invites for coffee chats

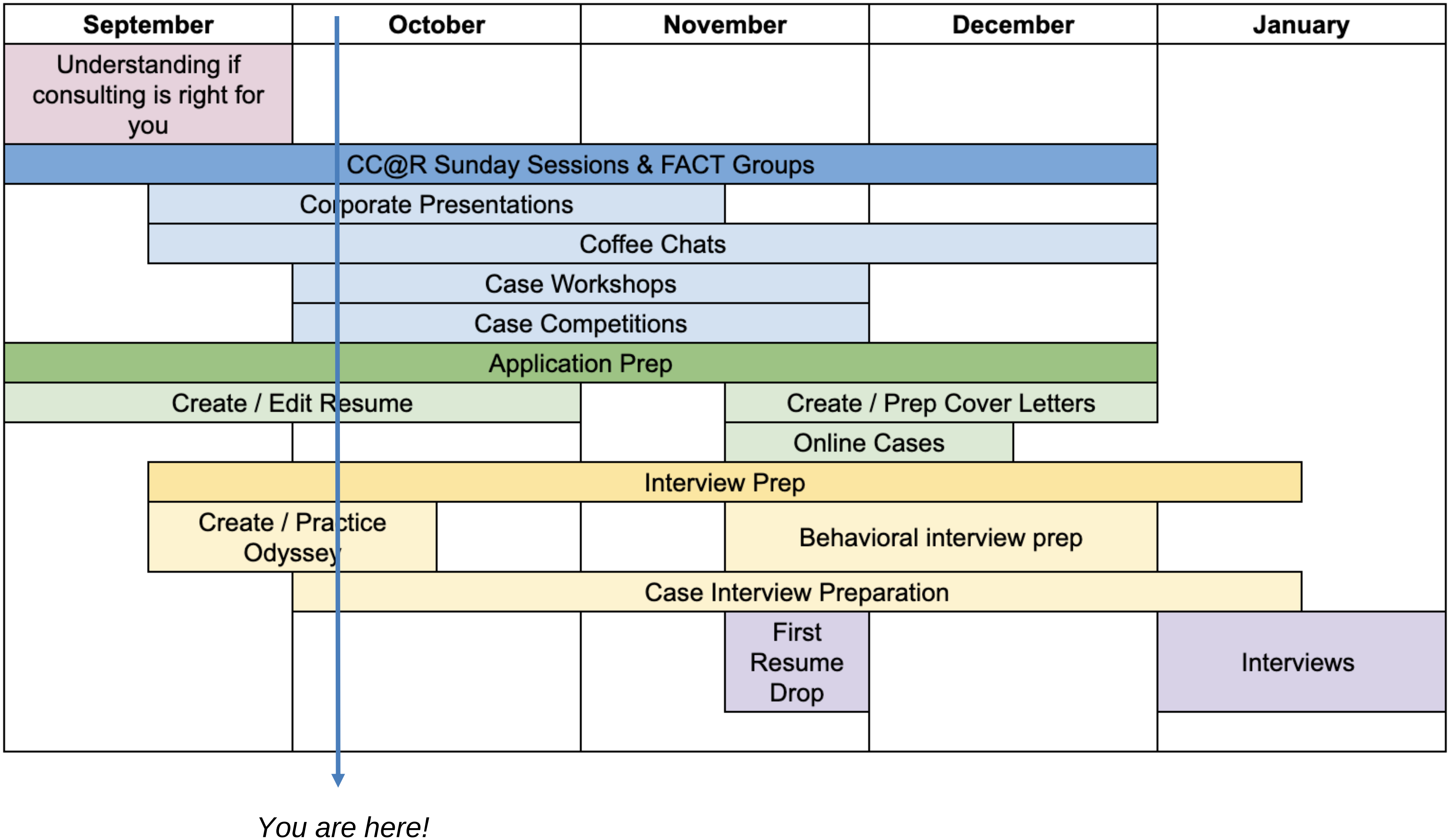
Do NOT no-show a coffee chat or meeting (even with an MBA2); MBA2's are plugged in with firms they interned at, and it can affect your candidacy for an interview

A few other notes

- **Do not request access to the CCR Shared Drive** - we will be adding new members (with view access) every week, and you will get access shortly if you have not gotten it already

- For our non-FTMBA members – CC@R is committed to support you with your recruitment journey, but we are not responsible for FACT groups and other CDO processes; **reach out to your respective career offices for more information on recruitment support!**

Big Picture: Recruiting Timeline



Our focus for today

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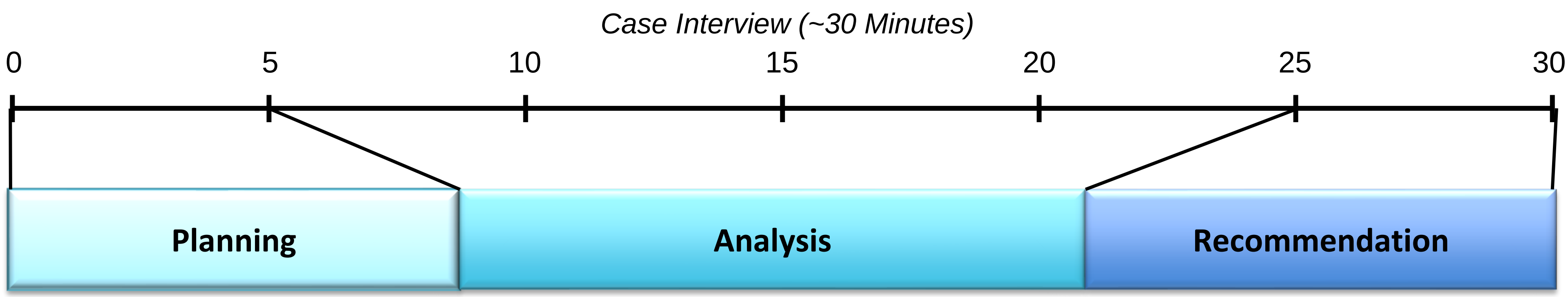

- **Dialogue:** Problem is stated and discussed / clarified (2-3 min). Usually, the interviewer has some more information to share.
- **Clarifying questions:** ask question to understand the goal
- **Framework:** Plan to conduct analysis and state hypotheses(~2 min)

- **Breaking down the problem**
  - **Hypothesis testing**
  - **Market sizing** (Case dependent)
  - **Charts / Exhibits**
  - **Brainstorming**

- Insights from analysis are synthesized into conclusions and recommendations
- Recommendations, risks, and next steps are delivered

Focus for today

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Lets revisit what we did last week...

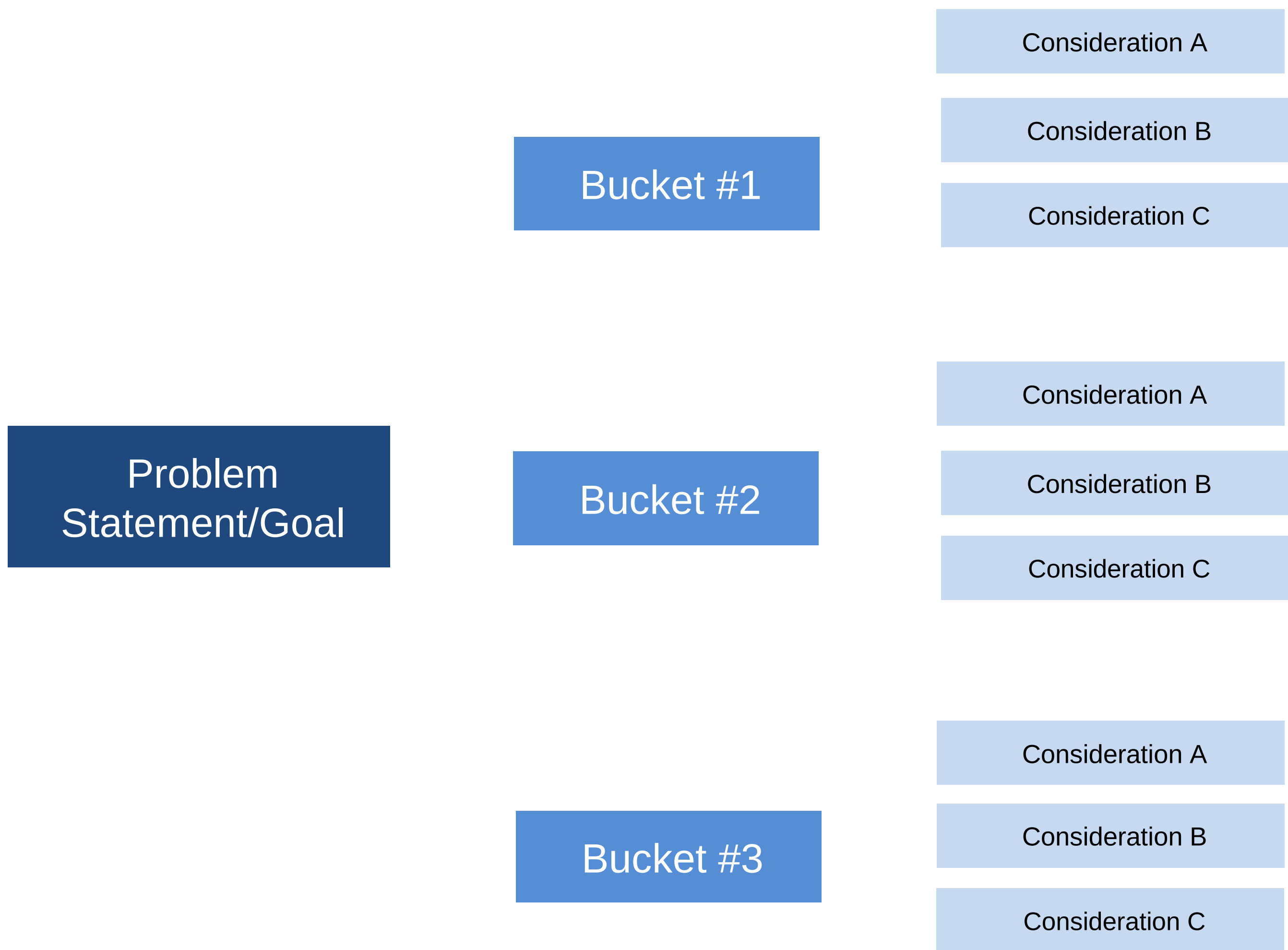
Last week, we watched Aman create a great framework to attack a business problem: How was Dairy Co. Going to expand nationally?

Let's think about what he did:

- ✓ **Listened to the case prompt carefully** and **took notes that worked for him** to breakdown the case
- ✓ Asked questions to **understand the client's goal** and any additional information he would need to craft his framework
- ✓ Asked the interviewer for a few minutes, then took his time crafting a **thoughtful, customized framework** based on his personal experience and expertise

What is a Framework?

The framework is your plan/ roadmap for how you're going to go about solving the case, this is generally what it should look like on your paper:



A framework should be

1. **Structured:** Have clear buckets and sub-buckets and be able to explain them succinctly
2. **Tailored:** Make it customized to the problem you're trying to solve
3. **"MECE" --> Mutually Exclusive Collectively Exhaustive**
   - Mutually Exclusive: No two buckets have the same sub-buckets
   - Collectively Exhaustive: Together, your framework covers all possible explanations
4. **Multi-layered**
   - Aim for 3's – 3 main buckets, 3 sub-buckets beneath each main bucket, *but this is NOT a rule*
5. **Creative:** Think outside the box, if time allows
6. **Hypothesis driven:** Have an idea of which bucket you think is the best solution to achieve the clients goal

Frameworks

Thinking about what you've learned in Econ, Strat, and in your past companies, **what frameworks do you already know**?

Some frameworks you *probably* already know

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3 C's: Company, Customers, Competition

Revenue = Profit + Costs

Costs = Fixed + Variable

SWOT: Strengths, Weaknesses, Opportunities, Threats

5 Forces: New Entrants, Substitutes, Competition, Customers, Suppliers

Organic "greenfield" vs In-organic growth

But don't be scared! Just remember:

1. You **already know** frameworks!
    - Pros & cons lists
    - Decision trees
    - How you decided to go to Ross
2. **Don't overthink this**
3. **Take your time at first**, speed will come over time
4. If you like a framework you did for a case type, write it down but **always be flexible**!

Let's watch another example

Our client is Wolverine Dairy, an Ann Arbor establishment selling ice cream, donuts, coffee and a limited selection of groceries. Known for their post-baseball game Ice Cream cones and Game Day donuts, WD has been a beloved Ann Arbor fixture for years. With local costs on the rise and an economic downturn looming, they need our help to stay afloat. Wolverine Dairy owners, lifelong Ann Arbor residents, have hired us to double their revenue.

Wolverine Dairy, Ross 2020

Frameworks: Tips & Tricks

- There is **no "right" framework**, but your framework needs to be **reasonably MECE** by avoiding obvious overlap and being thorough
- Good frameworks clearly **lay out all the questions you need to answer** to solve the case
- Framework will be the **guide throughout the entire case** – use it as a guide when you are stumped
- Bring in your **own relevant experience and interests** & use this to **show enthusiasm** when presenting the framework

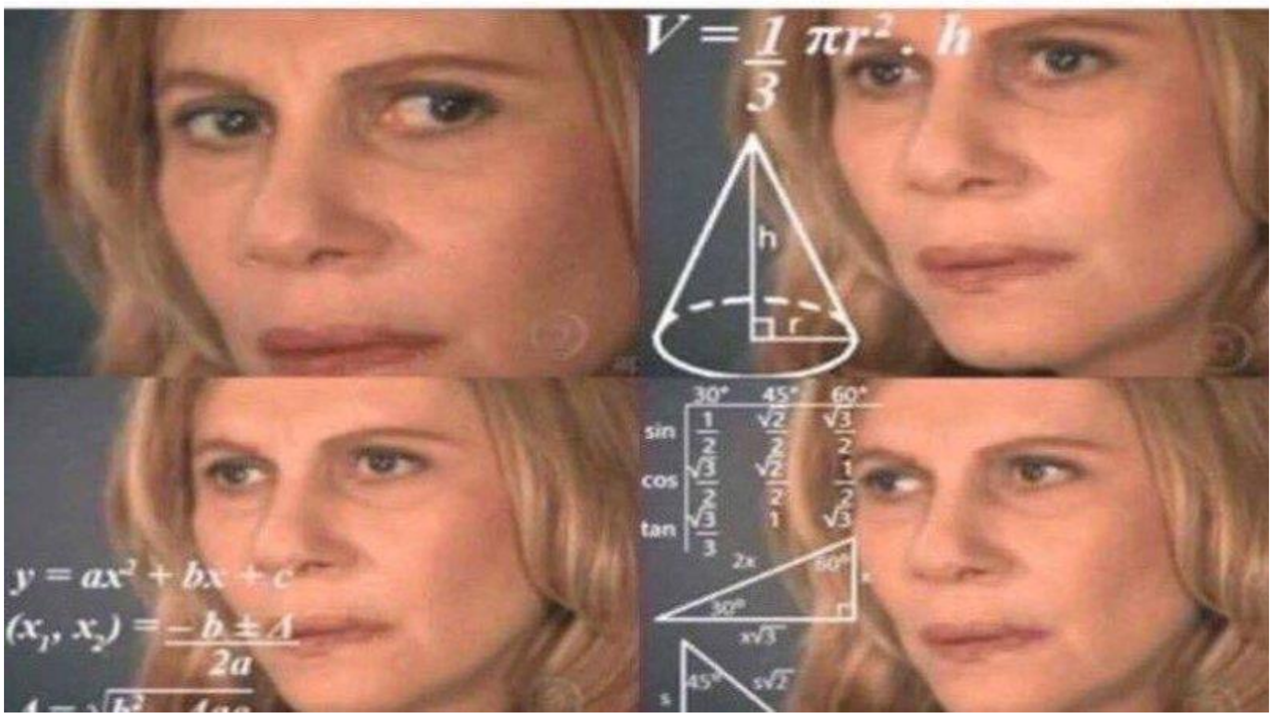
- Take your time at first, but eventually it should take you <2 mins to build out your framework
- Always start by presenting your big buckets first, then go back through the details of each bucket

Now it's your turn!

Our client is a private equity firm deciding whether to invest in a dairy farm in Iowa. The farm only produces milk. The client doesn't have any investments in the dairy industry or in farming more broadly. Should they purchase this dairy farm?

**Note: For most firms, the prompt will not be shown on screen*

Education Session #4
Part 2: Case Analysis



Which of the following are you most worried about when doing case math?

A. Quick Mental Math

B. Complicated calculations like NPV

C. Staying organized while doing math

D. Caser thinking I'm bad at math

Perceptions and Reality

If I make a mental math mistake, I'm done for!	Try to avoid mistakes, but it's absolutely not the make/break of an offer.
I should start explaining my thinking the moment I have an insight.	Take your time! Even if you already have an insight (be measured with your responses).
I must memorize NPV from finance and z-scores from statistics.	It's very unlikely that these will come up and understanding the concept > memorization.
The math will be harder IRL than in the case books.	Some cases in the books have "harder" math than any R2 interview I did.
If I get the "right" answer, I have completed the case and done my job.	Communication is more important than being "right" (though being right is good too). If you make a mistake, how you handle the recovery is very important

What Are They *Really* Looking For?

- **Structure**
  - Able to have a clear plan of action for how to use certain numbers and get to the right end goal
- **Comfort**
  - At ease, sanity check, cognitive aptitude, and a few tricks – able to connect numbers back to goal
- **Communication**: you're trying to tell a story
  - Might not be the "right" story, and that's OK!
  - Start with a high-level overview if you're stuck
- **Prioritization**
  - Various data sets: what's important (or not) and what are the connections?
- **Meaning Behind the Data / Relationships**
  - Business sense and recommendations
- **Precision**
  - Orders of magnitude, and potentially long(ish) multiplication/division
- **Speed**
  - Nobody has all day, but also not the #1 priority, especially at first

What is case math structuring – *an example*

Q. A baker plans to sell 200 loaves of bread for $6 each at the market. It costs the baker $2 to make each loaf. How much profit will they make?

When you do case math, you should be chasing the ***goal*** of the case.

1. Pause & make sense of the data – **don't just start calculating** something because you can
2. Talk through your plan **out loud**
3. Deal with the **pressure and complexity** (a lot of 0's, time, etc.)
4. Always put your answer in **context**, how does this tie to the goal of the case?

Some Common Formulas

Revenue = price per unit * # of units

Profit = Revenue - Costs

Markup = (Price – Cost) / Costs

Gross Margin = (Revenue – Cost) / Revenue

Net Margin = Net Income / Revenue

Return on Assets (ROA) = Net Income / Total Assets

Return on Investment (ROI) = Profit / Total $ invested

Breakeven = Investment Price/Fixed Cost - (Revenue – Variable Costs)

Rule of 72 = 72 / rate of return = # of years for invested capital to double

**Find more formulas in the Ross casebook*

TRICKS: Orders of Magnitude Under Pressure

Almost every single case uses large numbers, usually in the millions and billions; so many errors surround miscounted zeros which is entirely avoidable!

How? Take out the zeros!

- Use **K** for **Thousands** (three zeros) --> $4,000 = $4 K
- Use **M** for **Millions** (six zeros) --> $6,000,000 = $6 M (remember an M is also two K's) --> $6 K K
- Use **B** for **Billions** (nine zeros) --> $9,000,000,000 = $9 B (again, you can also use M K for billions)

TRICKS: Breaking down percentages

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When dealing with percentages, break them down into **percentages you know and can quickly calculate**:

What is 12% of 25M?

- 10% = 2.5M
- 1% = .25M * 2 = .5M
- 2.5M + .5M = **3M** --> *SANITY CHECK: does this sound roughly right? Well, if 2.5M is 10%, then we want just slightly more than that. Yes, 3M sounds right.*

What's 32% of 120B?

- 10% = 12B * 3 = 36B.
- 1% = 1.2B * 2 = 2.4B
- 36B + 2.4B = 38.4B --> *SANITY CHECK: think before you speak, does this sound right? Well, 50% of 120B is 60B, and 50% of 60B (or 25% of 120B) is 30B. Since 34% is slightly more than 25%, 38.4B should be right.*

Let's go back to Wolverine Dairy...

Our goal is to double revenue in 1 year

If Wolverine Dairy opens up a new location, they can expect the same annual revenue but 25% of sales at both locations will be cannibalized.

If we want to explore new offerings:

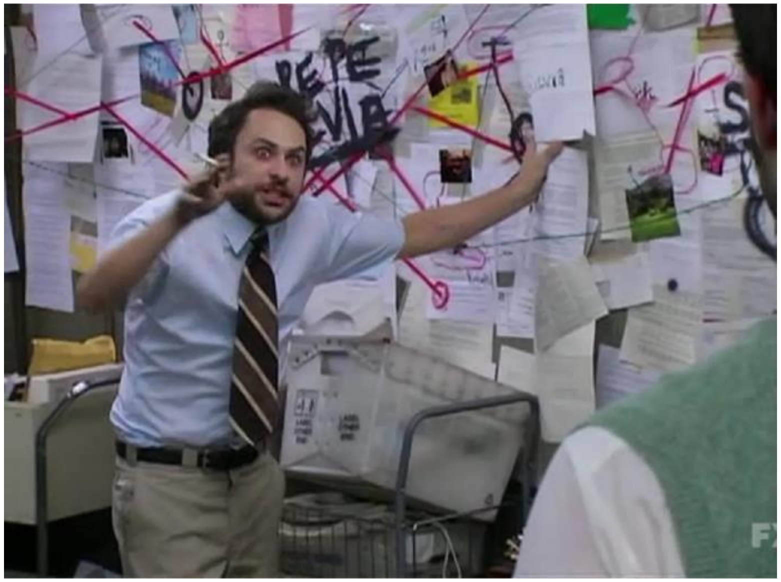
- Deli will earn $32.2k (new information)

And we can expect the categories to grow by the following amount:
- Grocery will grow by 10%
- Coffee will grow by 30%
- Ice cream grows by 16.67%

Current revenue by product line

Category	Annual Revenue
Coffee	$36k
Groceries	$30k
Ice Cream	$24k
Donuts	$10k

How many washers should Whirlpool sell at $2,000 per unit to generate $400 million in revenue?



Orders of Magnitude/Units Example 1

How many washers should whirlpool sell at \$2,000 per unit to generate \$400 million in revenue?

Step 1: Plan your approach

- Revenue = Price × Volume *so* **Volume = Revenue / Price**

Step 2: Simplify zeros

- Revenue = \$400,000,000 --> \$400 M
- Price = \$2 K / unit
- Volume = 400M / \$2K

Step 3: Calculate

- Divide: 400 / 2 = 200 and M (six zeros) / K (three zeros) = K (three zeros), so they need to sell 200K washers

Step 4: Gut check and contextualize!

- 200,000 washers seems like a lot for a small company but might not be that many for a gigantic one... does this number seem right to you?

And another example...

Your client's initial investment is $120M and the goal of the investment is to have a 20% return within 3 years.

What is the expected return per year?

Your client's initial investment is $120M and the goal of the investment is to have a 20% return within 3 years. What is the expected return per year?

Step 1: Plan your approach
Formula : ROI = Net Income / Investment --> ROI * Investment = Net Income

Step 2: Write out equation
- 20% (ROI) * 120M (Investment) = Net Income

Step 3: Calculate
- 20% * 120 = 24 M

TIP!
Convert % to fractions to quickly calculate:
- 20% = 1/5 --> 120 / 5 = 24

OR swap the decimals:
- 20% = .2 so we can remove one zero from 120 to make it 12 * 2 = 24

Step 4: Contextualize!
- Oh wait! The 24M net income is over all three years, and I need the return ***per year***. To get this I can divide the 24 M / year by 3 to get how much they'll return each year

Step 5: Calculate Again
- 24 M / 3 = **8 M / Year**

Think of the number in the context of the case

A simple way of understanding if your maths is incorrect or if you need to rethink the calculations a bit is to think of the number in the context of the case.

For example – if the client has an annual revenue of $100M and you calculated a potential revenue increase of $1B, does it really make sense?

Another example that Smarak saw last year in a final interview – a client wanted to break even on business expansion in <2 years, Smarak calculated a 17-year breakeven period because he miscalculated the net profit

What do I do if my math is wrong?

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1. If you recognize your mistake, it's OK! **Take a breath**, go back to the mistake and fix it. If it's a bigger mistake, **let your interviewer know** that you think something is a-miss.

2. If your interviewer catches your mistakes, that's also OK! They will be more concerned with how you handle the feedback and how you move forward than with the mistake itself. **Take the feedback, take a breath, and find the mistake**. It is not a make or break error!!

3. If you cannot find the mistake in your math, **don't get flustered.** Try again, and the interviewer might move on regardless.

4. If you messed up the math, **don't let it ruin the rest of your case.** You can still use the **brainstorming and recommendations to make a great impression!**

Remember, the goal is to be coachable!

Charts and Graphs

How to Approach Charts and Tables

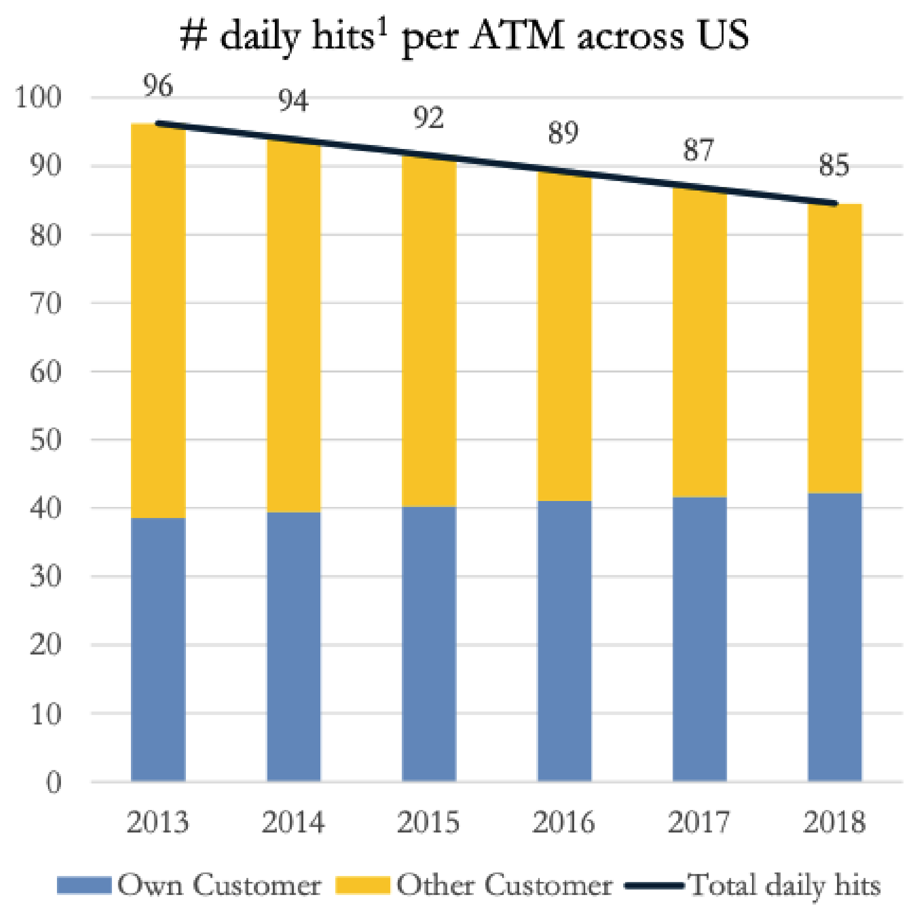
- **Keep in mind your objective**
  - You probably received the chart because you asked for some information, so look at it as a way to answer the question you had (aka test your hypothesis)
  - Be mindful of how the interviewer has framed the material - they will usually “introduce” the chart
- **Clear the chart**
  - Take a deep breath & orient yourself
  - Pay attention to: axes, title, legend, footnotes (carefully) - talk through what you're seeing and if you're unsure what a piece of info means, clarify
- **Engage in a dialogue to share your observations/insights**
  - Key insights/findings
  - Go beyond the obvious (second-level insights, calculation, connections, inter-dependencies)
  - Relate everything back to the original question and your original hypothesis
  - Propose next steps
  - Remember, they care more about how you communicate the findings than the findings themselves

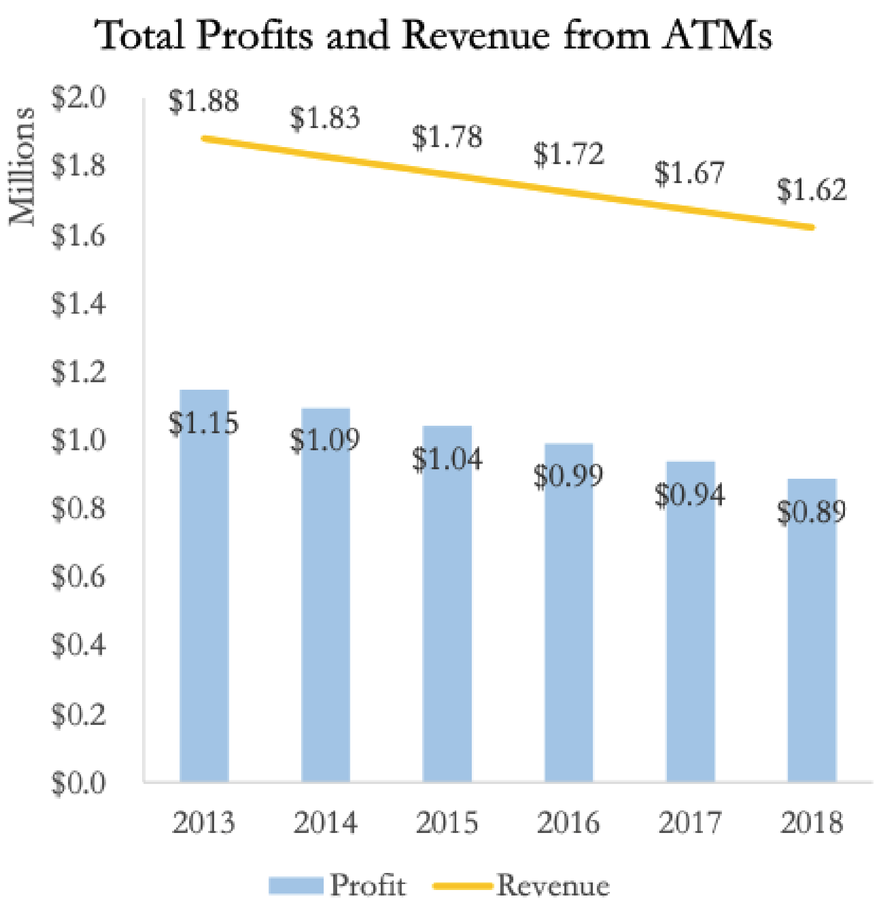
Example:

Case Prompt
Our client, American Bank, is a national retail bank operating in the US. ATMs have traditionally been a profitable channel, but the bank has started seeing declining operating profits from its ATMs. The CEO has hired your firm to help her analyze the reasons for this decline and solutions to improve usage. How would you approach this problem?

Example Chart #1

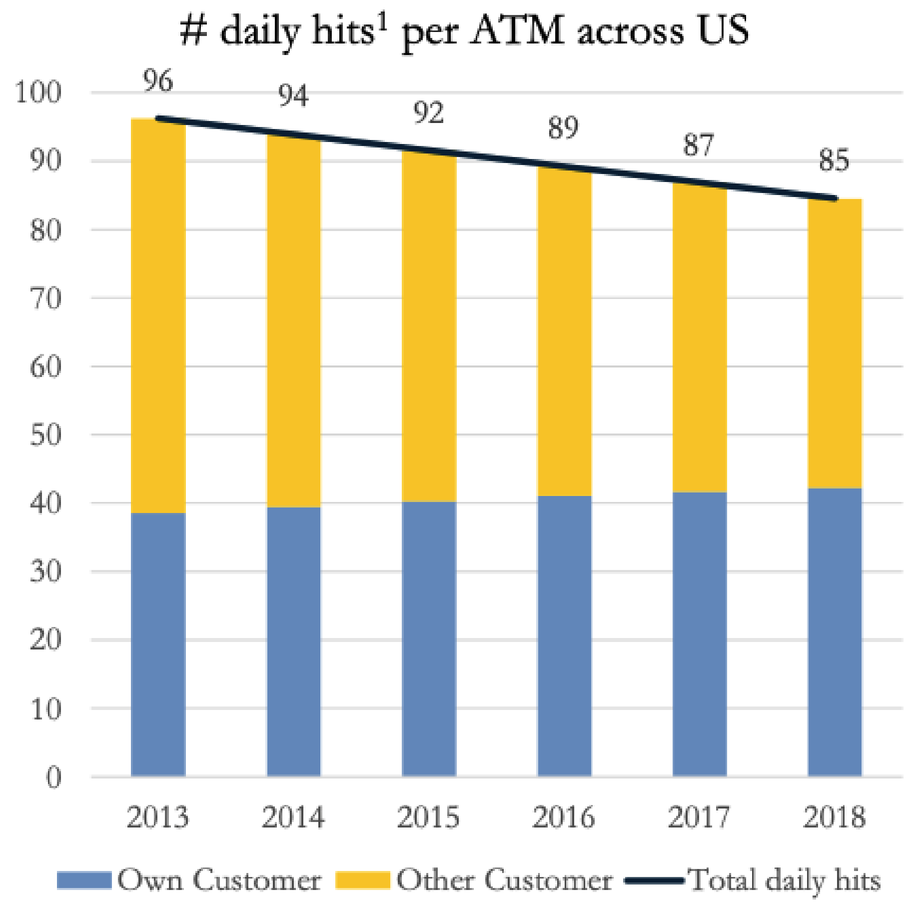
Exhibit 1 – American Bank ATM usage, profits and revenues in last 5 years





Note: Own customers are customers of American Bank, while Other customers are not customers of American Bank
1. hits = number of unique times an ATM was used (the word hits is used interchangeably with the word transactions)

Exhibit 1 – American Bank ATM usage, profits and revenues in last 5 years



1. # daily hits decreased YOY
2. # daily hits of other customers decreased

Revenue decreased

Next steps:

- Other customers – problem for further exploration
- Pricing model for own customers vs. other customers
- Costs

Note: Own customers are customers of American Bank, while Other customers are not customers of American Bank

1. hits = number of unique times an ATM was used (the word hits is used interchangeably with the word transactions)

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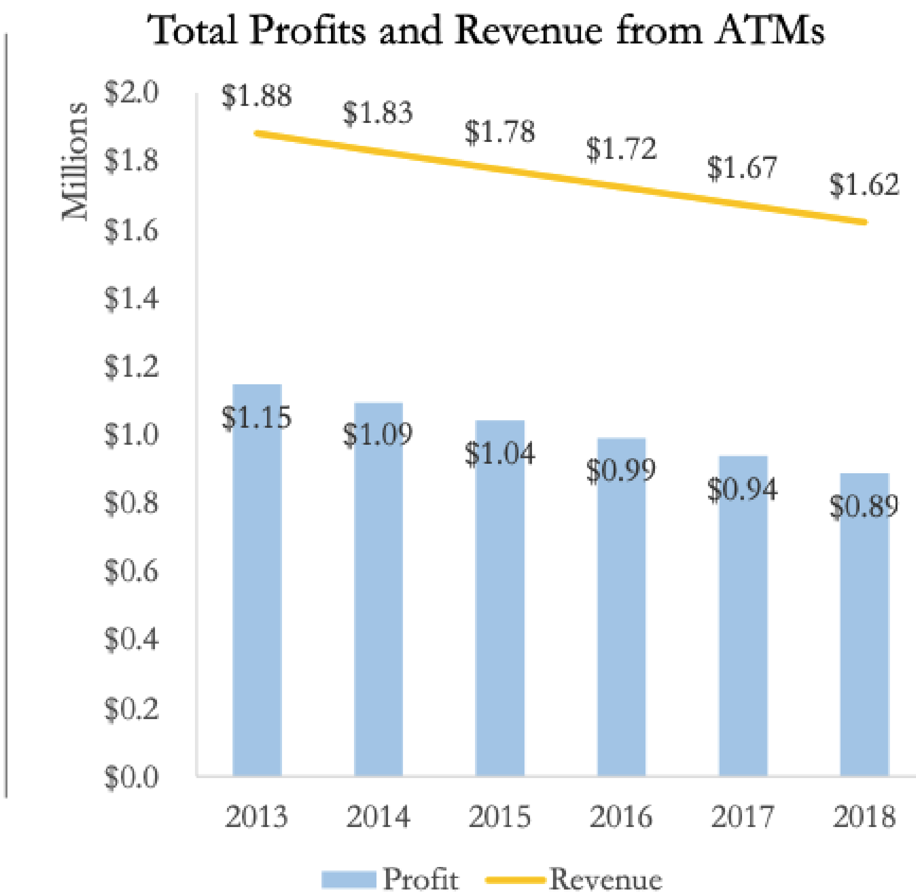
Exhibit 1 – American Bank ATM usage, profits and revenues in last 5 years

1. Revenue and profits decreased
2. Costs are constant

Focus on Revenue

Next steps:

- Other customers – problem for further exploration
- Pricing model for own customers vs. other customers
- ~~Costs~~



Note: Own customers are customers of American Bank, while Other customers are not customers of American Bank
1. hits = number of unique times an ATM was used (the word hits is used interchangeably with the word transactions)

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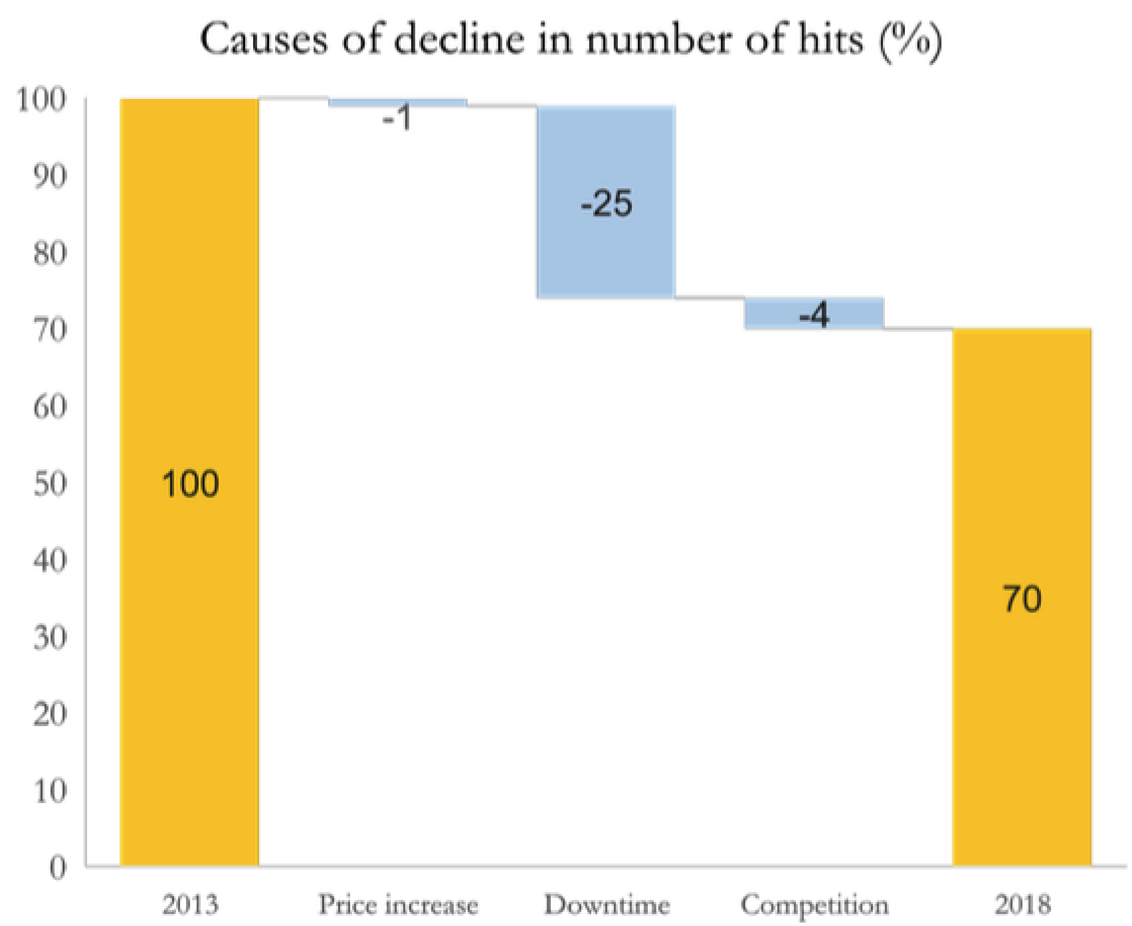
Remember to take time

- **Resist the temptation** to start talking right away!
- You **don’t need to have immediate insights**, and it can be a red flag even if you are technically “correct.”
- Take time to **structure your thoughts** against a general hypothesis (which should pertain to the prompt!).

Example Chart #2

Exhibit 4 – Reasons for reduction in hits



1. Vendor operated ATMs are ATMs where operations are outsourced to a vendor
2. American Bank cannot reduce transaction prices further from current levels
3. Uptime is the % of time the ATM is functional; Downtime = 100% - Uptime

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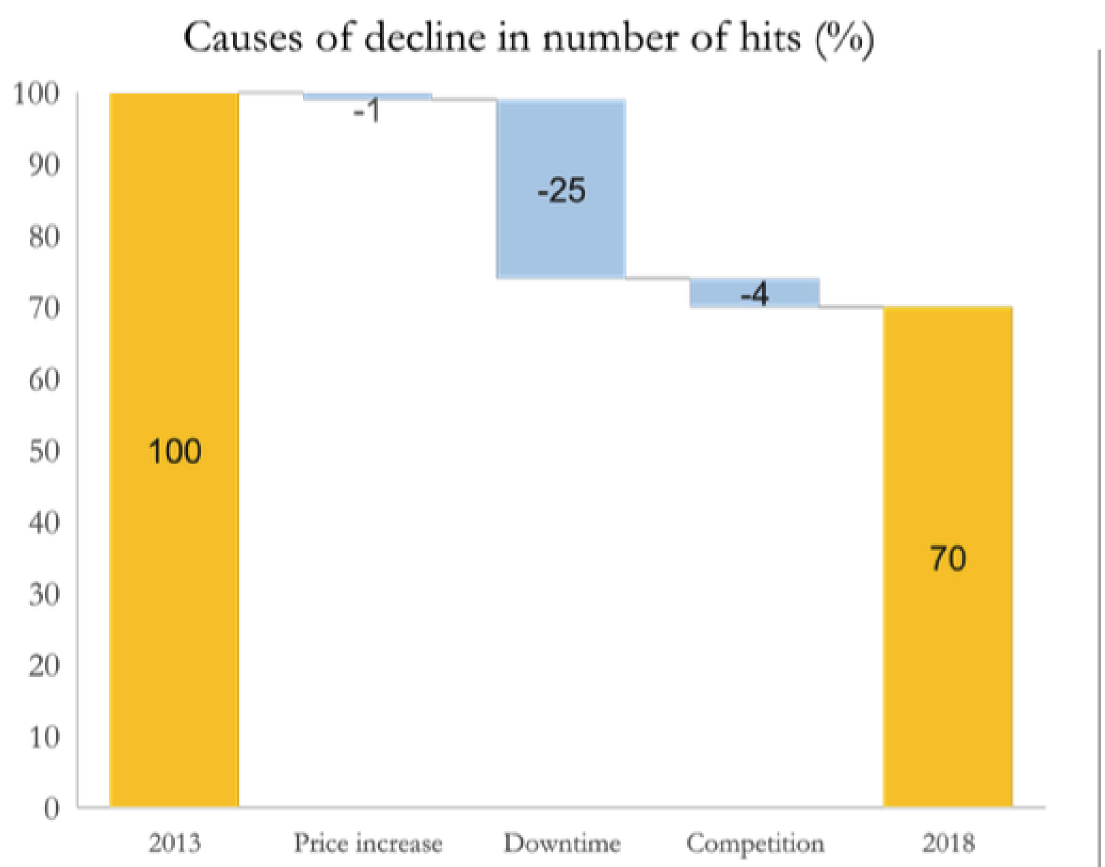
Exhibit 4 – Reasons for reduction in hits



- From 2013 to 2018, # of hits dropped 30%
- Three reasons contributed to the the decline
- Among them, Downtime is the major reason (25%)

Focus on Downtime

Next steps:

- Dig deeper into downtime (Downtime Causes)
- Quantify the opportunity

- Orient yourself to what is being built-up (or built-down)
- Pay attention to each bar's size/value relative to total size/value
- Pay attention to each label and how it ties back to the case

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Next Steps for Math and Chart Prep

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Math

- Practice Orders of Magnitudes, common percentages, and fractions, multiplication/ division tables
- Use Management Consulted to continue to improve math skills

Chart Analysis

- Practice with charts from old case books to pull out insights that you need and/or use **Management Consulted, Rocket Blocks, & McKinsey Chart of the Day**
- Practice taking 15-20s to organize thoughts and then present bucketed insights relevant to prompt

Deeper Insights

- Practice coming up with different hypotheses on same data - can you justify opposing narratives?
- Always be thinking critically: what was the prompt, and why might anyone care? Any risks in making assumptions?

Sign up to do your first full case with your FACT group leader!

Please take the pulse survey!

Appendix

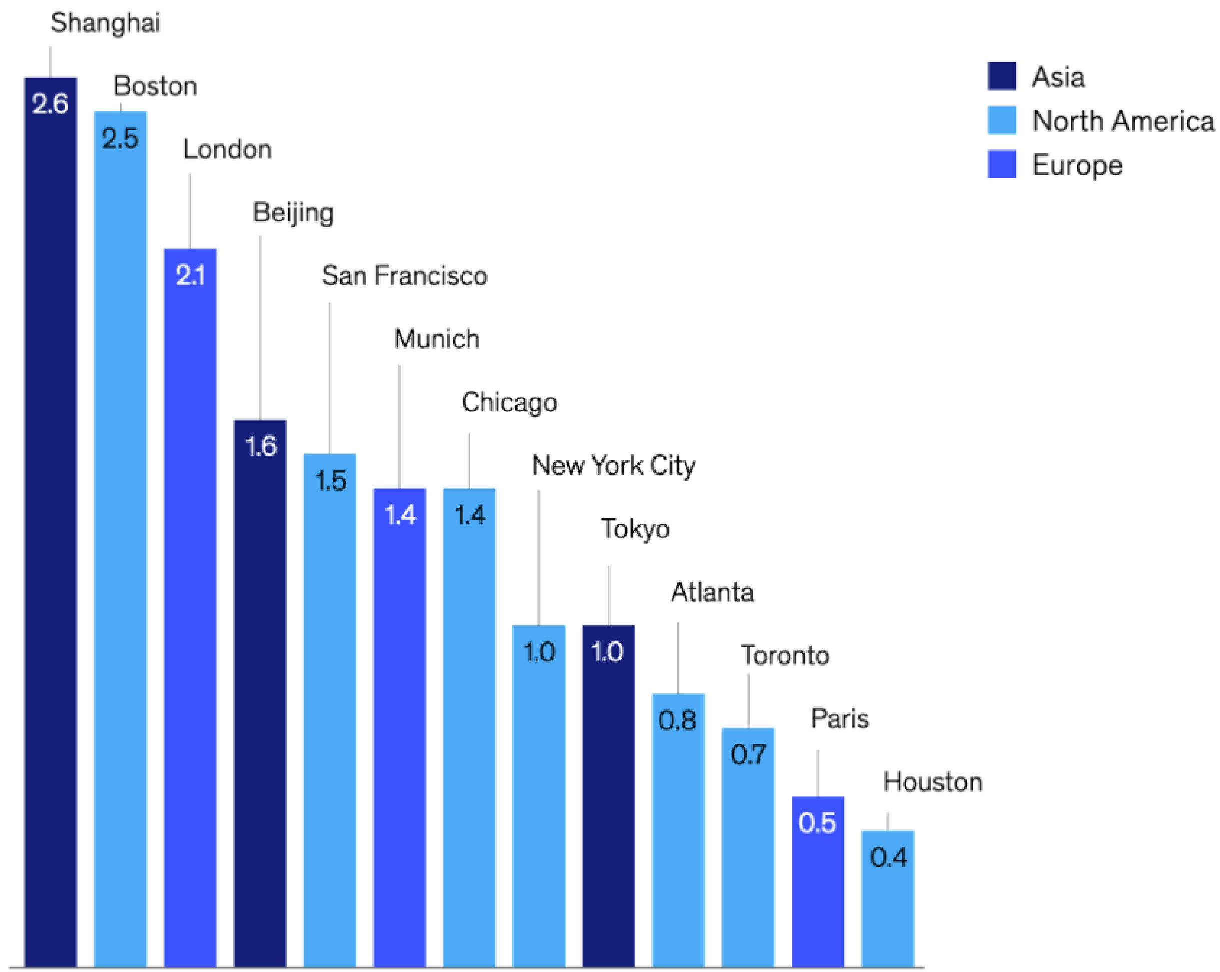
Let's continue the practice – build a framework

Your client is a wealthy former founder and CEO of a multi-national company interested in a new investment opportunity. The National Football League (NFL) wants to expand into Mexico by establishing an expansion franchise in Mexico City. This will be the NFL's first internationally based franchise, although the NFL has recently featured some games between American-based teams in Mexico City and London. The NFL is seeking owners for the team. Our client has sought our advice on whether they should pursue ownership.

NFL in Mexico (Darden'21)

Math Example – what are your key insights here?

Increase in housing stock if all excess office space were converted into residences by 2030,[1] %



[1]Excess office space means space projected to be vacant beyond what would result from the structural vacancy rate (defined as the average vacancy rate from 2014 to 2019).
Source: McKinsey Global Institute analysis